A. Research Proposal

1. Title

- Short, no waste words, no abbreviation, no full stop, should align with and reflect the objectives, method, and outcome of the study
- **Do not type in shaded areas, do not delete!**

2. Introduction (up to 250 words, citations 5-10 for the whole document, do not list out references here)

- Write in 'inverted triangle' format ideally in 3-paragraphs- 1) 'global, regional, and local' information about your study referring to the relevant literature; 2) What is known, controversies, and 3) Sum up with aim, rationale, and relevance of the study
- Citation in modified Vancouver, for details, see a recent issue of JPAHS
- Do not copy-paste the 'abstract' of a few randomly picked articles, avoid '**plagiarism**'
- **Do not type in the shaded areas, do not delete!**

Large language models (LLMs) are showing increasing utility in medicine by generating clinical summaries, interpreting and giving a diagnosis with management plan. They also provide educational resources for patients and healthcare professionals.[1] One of the major issues of these models is "hallucinations". It is seen when an LLM fabricates information instead of relying on valid evidence. While used in medical context, these hallucinations can include fabricated information and case details, invented research citations, or made-up disease details.[2] One of the convenient and accessible means for individuals to assess and understand the management of their mental health condition is by use of AI-driven tools and applications. However, the widespread trend of self-diagnosing mental health conditions and self-management along with checking the decision making of mental health experts through AI poses considerable risks.[3] If a healthcare professional as well as users are unaware of AI's limitations (i.e. AI hallucinations), this may cause harm to them due to inaccurate claims. On a user perspective simple change in prompts either accidental or deliberate can lead to AI hallucinations which shall then produce unsafe advices.

The measurement of the errors made by AI can help us to understand and prepare simple defenses and make us aware of the cost it carries when there is adversarial-hallucination in LLM. We aim to evaluate the occurrence and patterns of hallucinations by large language models (LLMs) when processing psychiatry vignettes derived from real inpatient cases at Patan Hospital.

3. Methods (details for reproducibility)

Detail enough for reproducibility, validity, and further extension

- Objectives- general, specific (number them e.g., 1,2,3…) do not use abbreviations in the objective, use measurable action verb;
- Study design, type of study, study site, study duration, sampling technique, sample size formula and calculation, study variables, inclusion and exclusion criteria, working definition;
- Procedure details-**who, where, when, how, whom**; data collection as per tools- user friendly pro forma and questionnaire (also in Nepali when relevant), dummy table(s) as per specific objectives (provide detailed tools at the end), data processing software, analysis tools; data management;
- Limitation of the study;
- Ethical consideration- Research participant safety, privacy, consideration for the vulnerable population, and conflict of interest.

- Provide citation where applicable, e.g., sample calculation
- **Do not type in the shaded areas, do not delete!**

General Objective

To evaluate the occurrence and patterns of hallucinations by large language models (LLMs) when processing psychiatry vignettes derived from real inpatient cases at Patan Hospital.

Specific Objectives:

1. To develop standardized psychiatry vignettes from de-identified inpatient records, each containing one deliberately fabricated clinical detail.
2. To test multiple LLMs under different conditions (default, safety-instruction, deterministic) and assess their responses.
3. To determine the hallucination rates of each LLM by model type, input length, and condition.
4. To compare whether shorter vignettes increase hallucination risk compared to longer versions.
5. To assess inter-rater reliability among psychiatrists in labeling model outputs.

Study design: Cross-sectional, experimental study

Study type: Analytical

Study site: Patan Academy of Health Sciences, Patan Hospital

Study duration: 6 months

Sampling technique: Purposive sampling

Sample size calculation:

Inclusion criteria:

- EMR of patients admitted to the psychiatry ward between January 2020 and December 2024.
- Admission records with complete data on key variables

Exclusion criteria:

- Admissions discharged within 24 hours (e.g., due to referral or absconding).
- Incomplete or missing data on major variables (e.g., detail work up, diagnostic formulations, diagnosis).

The de-identified psychiatry inpatient records from Patan Hospital will be collected after taking permission from the hospital director. All the inpatient data are electronically maintained in the hospital as a routine procedure. The diagnostic formulation and the diagnosis made by the team of psychiatry as a routine clinical procedure will be extracted and rewritten into standardized vignettes by consultant psychiatrist (researcher). Each vignette will include one intentionally fabricated clinical element (a non-existent test, drug, or symptom), while all other content remains realistic and clinically accurate. Two versions of each vignette will be created: short (50–60 words) and long (90–100 words). These vignettes will then be presented to different LLMs under three conditions: (i) default, (ii) safety-instruction, and (iii) deterministic (temperature = 0). Responses will be collected in structured format, and hallucinations will be defined as endorsement or elaboration of the fabricated detail. Two psychiatrists will independently code a subset of outputs for hallucination labeling, with inter-rater agreement quantified using Cohen's kappa. Descriptive and comparative statistics will be used to analyze hallucination rates by model, condition, and vignette length.

Ethical Considerations

The proposed study involves the use of retrospective electronic medical records (EMRs). The patient confidentiality will be strictly protected through data anonymization. Identifiable information will be removed prior to analysis, and only research team will access the dataset. As this is a retrospective study using de-identified data, a waiver of informed consent will be requested from the Institutional Review Committee (IRC), following ethical norms for minimal-risk research. The study will adhere to all institutional and national data protection regulations. The study will not influence real-time clinical care.

4. **List of cited references (total max 10)**

- Ideally, ¼th within 2 years, ¼ within 3-5 years, modified Vancouver with **at least one workable hyperlink** (details as JPAHS style, e.g. Scherer RW, Meerpohl JJ, Pfeifer N, Schmucker C, Schwarzer G, von Elm E. Full publication of results initially presented in abstracts. Cochrane Database Syst Rev. 2018;11:MR000005 | DOI | PubMed | Google Scholar | Full Text | Weblink |
- Cited references may be more than 10 for multicentric and funded research
- **Do not type in shaded areas, do not delete!**

1. Clusmann J, Kolbinger FR, Muti HS, Carrero ZI, Eckardt JN, Laleh NG, et al. The future landscape of large language models in medicine. Commun Med. 2023 Oct 10;3:141.

2. Maleki N, Padmanabhan B, Dutta K. AI Hallucinations: A Misnomer Worth Clarifying. In: 2024 IEEE Conference on Artificial Intelligence (CAI) [Internet]. 2024 [cited 2025 Sept 1]. p. 133–8. Available from: https://ieeexplore.ieee.org/abstract/document/10605268

3. Wimbarti S, Kairupan BHR, Tallei TE. Critical review of self-diagnosis of mental health conditions using artificial intelligence. Int J Ment Health Nurs. 2024;33(2):344–58.

5. Additional information - provide detail of all items *a-j* individually, write NA if not applicable

a. Budget- itemized breakdown (e.g., generic given below, modify as necessary), funding - detail for approval, or if you plan to apply for it. Assure that you will provide the document of approval and fee to IRC-PAHS
b. Work plan, Gantt chart
c. Data storage, sharing, dissemination details
d. Multicenter - details of study sites, detailed proposal, and approval from the parent proposal site
e. Foreign researcher (institutional affiliation)
f. Details about vulnerable populations, the sensitivity of local culture and social values
g. Bio-sample taking out/bringing in the country
h. Equipment/materials to bring in/out of the country
i. The detailed proposal if any in addition to this IRC-Form
j. Conflict of interests, if yes provide details
- Do not type in shaded areas, do not delete!

a. Budget (generic guideline, may modify as necessary)

SN	Particulars	Qty	Unit cost NRs	Subtotal NRs
1.	LLMs access for a month	3 (Paid Models) + 1or 2 (Free models as baseline)	GPT-type – 10,000 Claude-type - 10,000 Gemini-type - 5000 Open-source - 5000	30,000
2.	Internet service			
..				
..	Miscellaneous (unseen, justifiable, max. 10% of aggregate subtotal)			
			Total NRs	

b. 'Work plan, Gantt chart (generic guideline, may modify as necessary; **Use AD**)

SN	Particulars	Sept 2025	Oct/Nov 2025	Dec 2025	Jan-Feb 2025
1	Proposal development	■			
2	Data collection after ethical approval		■		
3	Data analysis/ thesis/article writing			■	
4	Final thesis/article submission				■

c. **Data storage, sharing, and dissemination details** (generic guideline, may modify as necessary):

During study- data will be stored in a password-protected computer of the researcher. After study- data will be deposited in the same computer for at least 5 years, with access to IRC-PAHS when required.

Provide details individually for each item, and write NA if not applicable for the following:

d. **Multicenter: NA**

e. **Foreign researcher (institutional affiliation): NA**

f. **Details about vulnerable populations: NA**

g. **Bio-sample taking out/bringing in the country: NA**

h. **Equipment/materials to bring in/out of the country: NA**

i. **The detailed proposal if any in addition to this IRC-Form: NA**

j. **Conflict of interests, if yes provide details: None**

B. Data Collection Tools / Pro forma

The Pro forma should contain the following contents:

- Title, Document ID number, Inclusion and Exclusion Criteria, Study variables, Data Collection Tools
- **Do not type in shaded areas, do not delete!**

C. Dummy Table:

List dummy table as per **EACH** specific objective
- Write the complete title of each dummy table to reflect the contents of the table
- **Do not type in the shaded areas, do not delete!**

Model	Condition	Short vignette words 50-100	Long vignette (n= <50)	Overall Hallucination Rate (%)
GPT-type	Default			
	Safety-instruction			
	Temp-0			
Claude-type	Default			
	Safety-instruction			
	Temp-0			
Gemini-type	Default			
	Safety-instruction			
	Temp-0			
Open-source	Default			
	Safety-instruction			
	Temp-0			

Dataset subset	Items rated (n)	Percent agreement (%)	Cohen's κ	95% CI for κ	Interpretation
Pilot set (all models)					
Main study (20% sample)					
Short vignettes only					
Long vignettes only					
Safety-instruction cond.					
Default condition					
Temp-0 condition					

Note: The researcher will be required to submit the copy of voucher (Laxmi Sunrise Bank, Account Number: 01811000295 in the name of PAHS) as required to the IRC-PAHS office as follows-

IRC Processing Fees

1. Non-funded research
   - *1.1 IRC will charge NRs. 1000/-*
   - *1.2 The processing fee for the research proposal of staff and/or students of PAHS will be waived*

2. Funded research
   - *2.1 For national funding, 2% of the site (PAHS) budget and not less than NRs. 3000/-*
   - *2.2 For international funding, 2% of the site (PAHS) budget and not less than NRs. 10,000/-*